Vivek Krishnan

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Education

2023–2027 **University of Pennsylvania**, Philadelphia, PA
B.A. in Physics and Mathematics, expected May 2027

Research Interests

My research interests lie in quantum gravity and holography, particularly the role of quantum entanglement in emergent spacetime geometry. I am interested in how semiclassical gravity and bulk locality arise from underlying quantum degrees of freedom through holographic duality and bulk reconstruction.

Research Experience

Feb 2026 – present **High-energy theory** — University of Pennsylvania
Undergraduate Researcher; senior thesis advisor: Jonathan J. Heckman

Investigating soft photon consistency relations for late-time QED correlators in de Sitter spacetime and their interpretation through generalized one-form symmetries. Developed Schwinger–Keldysh formulations of Ward identities in the Poincaré patch and examined the relation between late-time charge insertions and physical transverse soft-photon modes. Derived curved-spacetime effective descriptions of charged particles using heavy-quark effective theory (HQET) and soft-collinear effective theory (SCET). Current work examines whether soft gauge interactions factorize into Wilson lines, as required for extending the flat-space infrared symmetry construction.

Mar 2025 – May 2026 **Statistical physics and optimal inference** — University of Pennsylvania
Undergraduate Researcher; advisor: Andrea J. Liu

Developed a stochastic-thermodynamic framework for sequential inference using finite-state physical memories. Derived weak-signal diffusion limits for likelihood-ratio updates and a Green–Kubo friction tensor for reversible Markov memory dynamics, yielding a dissipation metric on the probability simplex. Formulated optimal update protocols under accuracy and sampling-time constraints using stochastic control, Feynman–Kac equations, and constrained optimization. Generalized the framework to n memory states and j hidden states. Manuscript preparation continues with Adam Gabriel Frim and Andrea J. Liu.

Apr 2024 – Mar 2025 **Computational particle physics** — University of Pennsylvania
Undergraduate Researcher; advisor: Dylan S. Rankin

Developed a vector-quantized variational autoencoder (VQ-VAE) for unsupervised anomaly detection in ATLAS particle-collision data. Trained classifiers directly on quantized latent representations to test signal retention and identify anomalous events without decoder-based reconstruction of the original detector data.

Jan 2022 – Mar 2024 **Computational XFEL research** — Arizona State University
Student Researcher; advisor: Richard A. Kirian

Developed an automated Python pipeline for characterizing gas dynamic virtual nozzles (GDVNs), including a machine-learning classifier for drip–jet regimes (99.3% accuracy). Contributed particle image velocimetry and statistical analyses of liquid microjets for X-ray free-electron laser experiments.

Publications and Manuscripts

Manuscript	**V. Krishnan**, A. G. Frim, and A. J. Liu. "Optimal Statistical Inference Under Thermodynamic Constraints." *Manuscript in preparation.*
Nov 2024	K. Karpos, S. Zaare, D. Manatou, R. C. Alvarez, **V. Krishnan**, C. Ottmar, J. Gilletti, A. Pableo, D. Doppler, A. Ansari, R. Nazari, A. Ros, and R. A. Kirian. "Comprehensive characterization of gas dynamic virtual nozzles for x-ray free-electron laser experiments." *Structural Dynamics* **11**, 064302 (2024). doi:10.1063/4.0000262

Presentations

Mar 2026	**Optimal Finite-Time Statistical Inference** Contributed oral presentation, APS Global Physics Summit, Denver, CO.

Selected Advanced Coursework

Graduate physics	Quantum Field Theory I–II; Quantum Mechanics I–II; Statistical Mechanics; General Relativity; Electromagnetic Phenomena; Mathematical Methods of Physics; Advanced Topics in Theoretical Physics.
Mathematics	Advanced Analysis; Differential Geometry; Geometric Analysis and Topology; Abstract Algebra; Advanced Linear Algebra; Complex Analysis; Partial Differential Equations.
Spring 2026	**Supersymmetry and Supergravity — Independent study** With Mirjam Cvetič; supersymmetry algebras, superfield formulations, and local supersymmetry in supergravity.

Teaching Experience

Teaching Assistant, University of Pennsylvania

Fall 2026	PHYS 5503 — General Relativity
Spring 2027	PHYS 5516 — Electromagnetic Phenomena *(upcoming)*

Honors and Research Funding

Roy and Diana Vagelos MLS Scholar
Received $20,000 in summer research funding across 2025 and 2026.

Academic Service

2025–present	**President**, Penn Society of Physics Students Organize faculty talks, graduate student Q&A panels, and social/academic events fostering interaction within Penn's physics community.
2026–present	**Research Peer Advisor**, Center for Undergraduate Research and Fellowships Advise a cohort of first- and second-year students on identifying research opportunities, contacting faculty, and selecting projects.
2023-present	**Problem Writer**, Penn Math Contest Contribute problems for the inaugural competition serving high school students across the Greater Philadelphia area.